```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate a population (e.g., normal distribution)
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)

# Step 2: Random sampling
sample_sizes = [30, 50, 100] # different sample sizes to consider
num_samples = 1000 # number of samples for each sample size

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sample_means = {}

for size in sample_sizes:
sample_means[size] = []
for _ in range(num_samples):
sample = np.random.choice(population, size=size, replace=False)
sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
plt.subplot(len(sample_sizes), 1, i+1)
plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Population
Mean')
plt.title(f'Sampling Distribution (Sample Size {size})')
plt.xlabel('Sample Mean')
plt.ylabel('Frequency')
plt.legend()

plt.tight_layout()
plt.show()
```
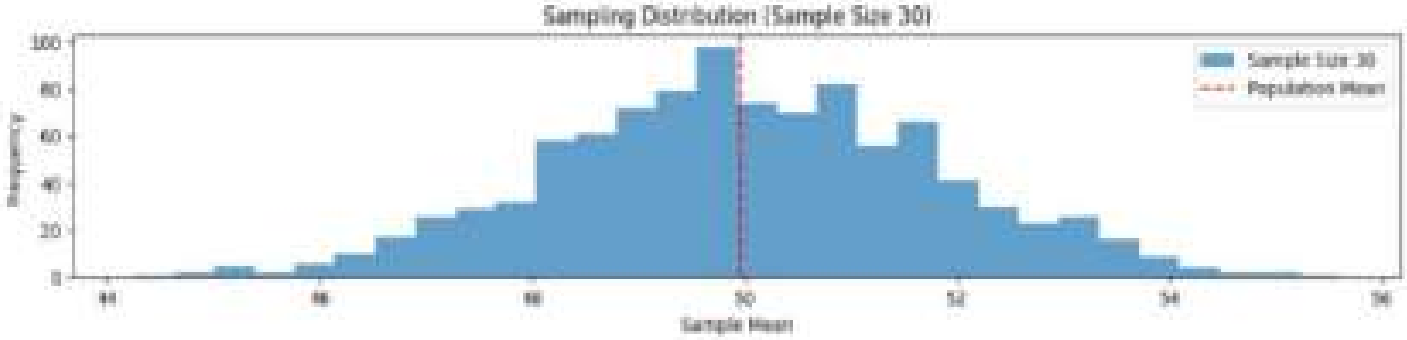
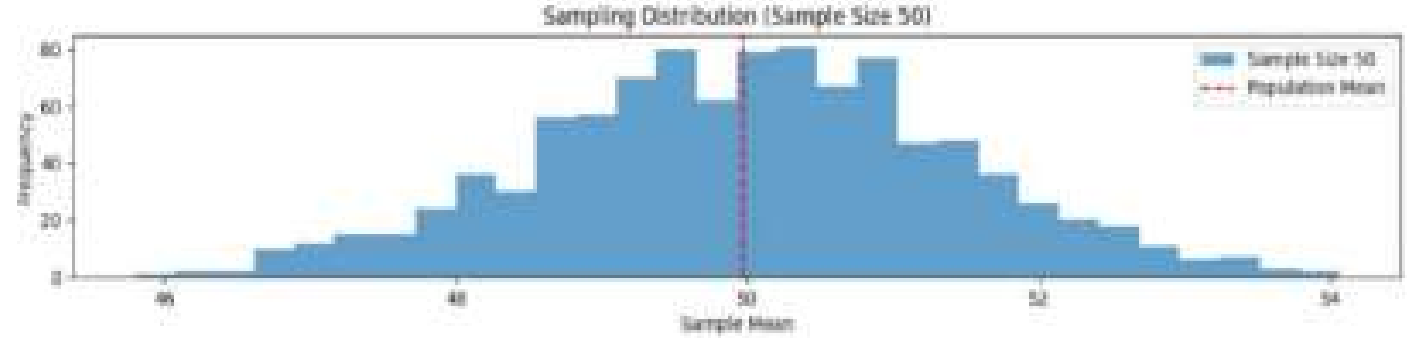
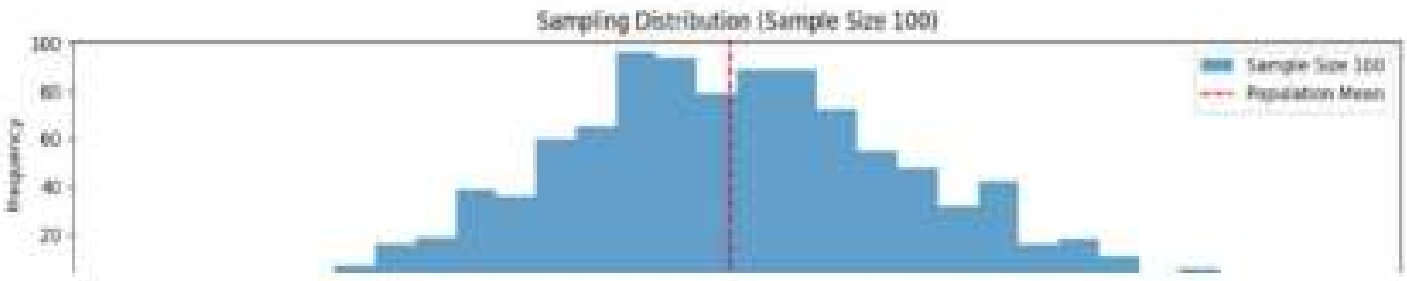


Sampling Distribution (Sample Size 30)
Frequency
Sample Mean
Sample Size 30
Population Mean


Sampling Distribution (Sample Size 50)
Frequency
Sample Mean
Sample Size 50
Population Mean


Sampling Distribution (Sample Size 100)
Frequency
Sample Size 100
Population Mean